ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ROBOTICS AND INDUSTRIAL AUTOMATION

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Robotics and Industrial Automation paper 1 · practice a.

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How to mark: each ✓ is one mark. Accept any standard ladder notation and addressing. Answers must relate to the case study where asked. Accept other correct answers.

SECTION A

QUESTION 1

1.1 D ✓
1.2 B ✓
1.3 D ✓
1.4 B ✓
1.5 C ✓
1.6 A ✓
1.7 B ✓
1.8 D ✓
1.9 A ✓
1.10 C ✓

(10 × 1) [10]

QUESTION 2

2.1 True ✓
2.2 True ✓
2.3 False ✓
2.4 True ✓
2.5 True ✓
2.6 True ✓
2.7 False ✓
2.8 True ✓
2.9 True ✓
2.10 False ✓

(10 × 1) [10]

QUESTION 3

3.1 A ✓
3.2 D ✓
3.3 F ✓
3.4 E ✓
3.5 I ✓
3.6 J ✓
3.7 G ✓

3.8 C ✓

3.9 H ✓

3.10 B ✓

(10 × 1) [10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1 Any FIVE, name ✓ + function ✓: CPU – executes the program; power supply – provides power (usually 24 V DC) to the PLC; input module – receives signals from sensors and switches; output module – switches motors, valves and lamps; memory – stores the program and data; programming device / software – used to write and download the program; communication module – connects to HMI, robot, network (10)

4.2 Any FOUR: easy to change the program without rewiring ✓; fewer wires and less panel space ✓; faster fault-finding (monitoring, diagnostics) ✓; reliable – no mechanical relay wear ✓; timers and counters built in; can communicate with HMIs and SCADA; cheaper for complex systems (4)

4.3 (a) input, digital ✓ (b) input, analogue ✓ (c) output, digital ✓ (d) output, analogue ✓ (4)

[18]

QUESTION 5

5.1.1 It is a latching (seal-in / holding) contact ✓ – it keeps the motor running after START is released ✓. (2)

5.1.2 Pressing START energises Q0.0 and the conveyor motor starts ✓; Q0.0 latches ✓; the Q0.0 contact in rung 2 starts timer T1 ✓; after 10 s, T1's contact closes and the warning lamp Q0.1 switches on ✓. (4)

5.1.3 The motor stops (latch is broken) ✓; T1 resets so the lamp switches off ✓. (2)

5.1.4 Fail-safe: if the wire breaks or the contact fails, the input goes OFF and the motor stops ✓, so a broken stop circuit cannot prevent stopping ✓. (2)

5.2 Rung 1: I0.3 (NO) → CTU C1 count input with PV = 24 ✓✓; I0.4 (NO) → reset input of C1 ✓✓; rung 2: C1 contact (NO) → Q0.2 coil ✓✓ (6)

5.3 START buttons I1.0 and I1.1 in parallel ✓✓, with Q0.3 latching contact also in parallel with them ✓; STOP inputs I1.2 and I1.3 (NO contacts in the program, since the buttons are NC in the field) in series ✓✓; Q0.3 coil ✓ (6)

[22]

QUESTION 6

6.1 Any FIVE with function ✓✓✓✓✓: six-axis robot – picks and places packets; robot controller and teach pendant – runs and programs the robot; end effector (vacuum gripper) – holds the packets; conveyor – brings packets; sensors (photo-eye / vision) – detect packet position; PLC – controls the conveyor and coordinates the cell; safety fence, interlocked door, light curtain – protect people; HMI – operator control (5)

6.2 Any FOUR, each described ✓✓: interlocked guard door that stops the robot when opened; robot switched to manual / reduced-speed mode with an enabling (dead-man) switch on the pendant; lock-out / tag-out of power and air before clearing jams; emergency stop buttons inside and outside the cell; light curtains / safety mats at openings; trained and authorised operators only; risk assessment and safe work procedure (8)

6.3 The PLC sets an output "packet ready" when the sensor detects a packet in position ✓; the robot controller reads this signal and starts picking, then sends "pick complete" back to the PLC ✓; the PLC then moves the conveyor to the next packet – each step waits for the other's confirmation ✓. (3)

[16]

QUESTION 7

7.1 An HMI is a local operator panel (touch screen) for one machine or cell ✓✓; SCADA is software on a central computer that supervises, controls and records data from many machines, PLCs or sites ✓✓. (4)

7.2 Any FIVE ✓✓✓✓✓: running / stopped status; number of packets / boxes completed; conveyor speed; alarms and faults (e.g. gripper vacuum lost, door open); emergency stop status; production rate per hour; robot mode (auto/manual); start, stop and reset buttons (5)

7.3 Any THREE: sensors and data analysis predict breakdowns (predictive maintenance) ✓; real-time production data helps managers improve efficiency ✓; machine vision checks quality automatically ✓; remote monitoring; digital twin to test changes; automatic ordering of materials (3)

7.4 Any TWO risks with a solution (½ + ½ each): hacking / cyber-attacks – firewalls, network segmentation ✓; data theft – encryption, strong passwords; malware stopping production – updates, backups, antivirus ✓ (2)

[14]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE